

Junming Zhao

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Education

University of New South Wales (UNSW) | Sydney, Australia

2024–Present

PhD Candidate in Computer Science and Engineering

- Formalising a multithreaded network subsystem on the seL4 secure microkernel using rely-guarantee reasoning and separation logic
- Supervisors: Dr. Rob Sison, Dr. Thomas Sewell, and Dr. Miki Tanaka
- Research Team: UNSW Trustworthy Systems Team, led by Prof. Gernot Heiser

The Australian National University (ANU) | Canberra, Australia

2018–2023

Bachelor of Advanced Computing (Honours) and Bachelor of Mathematical Sciences

- University Medal
- First Class Honours
- GPA: 6.9/7.0

Selected Research Outputs

Papers and Preprints

- Samuel Tyler, Alessandro Legnani, **Junming Zhao**, Zhewen Shen, Rihui Wu, Miki Tanaka, and Gernot Heiser. *SMT-based Deductive Verification of Device Drivers using the Pancake-to-Viper Transpiler*. To appear in Proceedings of the 14th Workshop on Programming Languages and Operating Systems (PLOS 2026). [\[Paper\]](#)
- **Junming Zhao**. *Mechanising Local Rely-Guarantee Reasoning over the seL4 Trace Monad*. Workshop paper, Isabelle Workshop 2026, held as part of the Federated Logic Conference (FLoC 2026), July 2026. [\[Paper\]](#) [\[Slides\]](#)
- **Junming Zhao** et al. *Verifying Device Drivers with Pancake*. arXiv:2501.08249 [cs.PL], 2025. [\[Preprint\]](#)

Selected Talks

- *A Rely-Guarantee + Separation Logic for Verifying Microkit-Based Systems*. Formal Methods Australia Workshop (FMOz 2026), June 2026. [\[Slides\]](#)
- *Verifying Device Drivers with Pancake*. seL4 Summit 2025, September 2025. [\[Video\]](#) [\[Slides\]](#)
- *Using MMTk Garbage Collection in GHC*. Lightning talk, Haskell Implementors' Workshop (HIW 2023), co-located with ICFP 2023, September 2023.

Research and Professional Experience

Summer Research Fellow

2026

Renaissance Philanthropy AI for Math Summer Fellowship

- Mechanising angelic nondeterminism in Isabelle/HOL
- Research in parallelism for angelic processes
- Supervisor: Dr. Pedro Ribeiro

Research Engineer

2023–2024

UNSW Trustworthy Systems Team

- Implemented device drivers in Pancake
- Benchmarked network performance across different hardware platforms and user-level networking policies
- Deployed systems on cloud platforms

Research Assistant

2022–2024

ANU MMTk Team

- Ported MMTk (Memory Management Toolkit) to the GHC (Glasgow Haskell Compiler) runtime as part of Google Summer of Code 2022
- Introduced flexible garbage-collection plans in GHC and extended MMTk to better support runtimes for functional languages
- Supervisors: Prof. Steve Blackburn and Ben Gamari

Software Engineering Intern

2021

Rexergy, Canberra

- Developed Python models for household thermal appliances
- Performed linear and nonlinear optimisation using Gurobi, CBC, and Pyomo

Teaching Assistant and Education Developer

2020–2023

ANU

- Facilitated laboratory sessions and tutorials across undergraduate and postgraduate courses in computer systems, mathematics, machine learning, computer vision, and software engineering
- Developed and graded written and programming assignments

Awards and Scholarships

- seL4 Software System Scholarship (2024)
- ANU College of Engineering, Computing and Cybernetics Northrop Grumman Scholarship (2023)
- ANU Summer Research Scholarship (2021, 2022)
- ANU Dean's List Honouree with Chancellor's Letter of Commendation (2020, 2021)
- ANU Terrell International Scholarship (2019)